

Vitor Negromonte Cabral de Oliveira

Last update: May 22, 2026

Education

Federal University of Pernambuco

Recife

B.Sc Information Systems

2022–2028

Originally enrolled in Statistics (2022–2025), then transferred to Information Systems. Activities: Artificial Intelligence League co-founder, Teaching Assistant (Introduction to Deep Learning, Computational Creativity), Research Assistant.

Work Experience

Applied Machine Learning Engineer (FACEPE) – part-time

Dec 2025 – Present

Sinapse - Polícia Civil de Pernambuco

Recife, BR

Responsible for end-to-end criminal intelligence backend, including architecture, data platform, and governance.

- **System Architecture:** Designed a modular, event-driven monolith to decouple AI workloads and deliver sub-second API performance.
- **Data Platform:** Built asynchronous ETL pipelines following Medallion Architecture, leveraging MinIO for immutable storage and PostgreSQL for structured investigative data.
- **Governance:** Designed WORM-compliant auditing and automated PII de-identification to ensure LGPD compliance.
- **Model Evaluation:** Design evaluation pipelines for OCR and LLM models, measuring accuracy, latency, and domain relevance.

Stack: Python (FastAPI), PostgreSQL (pgvector), MinIO, Docker, Redis, Celery.

Machine Learning Engineer

Jan 2025 – Sep 2025

Freelancer

Remote

Designed and deployed a full-stack computer vision pipeline for agricultural applications. Managed data collection, dataset creation, image preprocessing, model training, and real-time edge inference.

Stack: Python (PyTorch, OpenCV), Docker, Linux, Shell, Git.

Data Scientist

Nov 2023 – Jul 2024

redduo.ai

Recife, BR

Supported business intelligence projects through data analysis and automation tool development. Developed and optimized an NLP pipeline, improving model performance and enabling scalable deployment across production environments.

Stack: Python (Scikit-learn, PyTorch), AWS (Lambda, Cognito), Power BI.

Research Experience

Center for Informatics, Federal University of Pernambuco

Dec 2025 – Present

Undergraduate Researcher

Recife, BR

Supervisor: Cleber Zanchettin. Developed a mechanistic interpretability framework for recursive neural architectures. Analysed dynamical systems properties (Jacobian spectra, attractor basins) and investigated how training objectives shape algorithmic reasoning compared to Transformers.

National Institute of Science and Technology in Software Engineering (INES)

Aug 2023 – May 2024

Undergraduate Research Assistant

Remote/Recife, BR

Assisted in the development of quantitative analysis tools and contributed to improving accessibility features in mobile applications aimed at supporting adults on the autism spectrum. Published at SBSI – 2025 and IHC 2024.

Extracurricular

President & Co-founder (Former Outreach Director)

Mar 2024 – Present

Ligia – Federal University of Pernambuco's Artificial Intelligence Academic League

Recife, BR

Co-founded and currently lead an AI research group affiliated with CIn-UFPE. Direct a team of ~30 researchers across applied NLP, Machine Learning, and Computer Vision tracks. Previously, as Outreach Director, led external relations to establish strategic partnerships within the public and private sectors and organized technical events to integrate academic research with the local tech ecosystem.

Co-organizer

Nov 2025

II Advanced Summer School on Responsible AutoML

Recife, BR

Event made by the Center for Informatics (UFPE) and the Engineering Faculty of the University of Porto. Coordinated outreach and partnership with interested organizations.

Projects

F1Predict: Formula 1 Race Prediction System – [🔗](#)

Built an end-to-end system for predicting Formula 1 race outcomes, including data collection, feature engineering, model training, and deployment. Developed a REST API with FastAPI and PostgreSQL for data management and model serving. Achieved a 1-second prediction error, demonstrating high real-time accuracy. Tools: Python, PyTorch, scikit-learn, FastAPI, PostgreSQL.

MARS: Multi Agent Recommendation System – [🔗](#)

Developed a FastAPI-based system that automates research paper collection from ArXiv using LLM agents to review and filter relevant papers. Tools: Python (FastAPI).

Parkinson Diagnosis using Computer Vision: Campus Party'24 Keynote – [🔗](#)

Created a CNN model for Parkinson's disease detection using spiral-drawing images. Outperformed state-of-the-art methods by $\approx 10\%$, achieving 95% accuracy on the benchmark dataset. Tools: Python, PyTorch, OpenCV, Optuna, Zeus.

Computer Vision in Breast Cancer Diagnosis: A Comparative Study with CBIS-DDSM Data – [🔗](#)

Performed a comparative study of multiple CNN architectures for early breast cancer detection using mammography images. Tools: Python, TensorFlow, OpenCV.

Publications

- [2] D. M. Ribeiro, F. V. Melo, **V. Negromonte**, C. Pereira, A. P. C. Steinmacher, K. Gama, "A Mapping Review to Understand Web and Mobile Apps Accessibility for Adults with Autism", SBSI – 2025
- [1] D. M. Ribeiro, F. V. Melo, **V. Negromonte**, G. W. Matias, A. Farias, C. Azul, A. P. Chaves, K. Gama, "A Comparative Study on Accessibility for Autistic Individuals with Urban Mobility Apps", IHC – 2024

Awards & Grants

Cohere Labs Catalyst Grants Program

2025

Teaching

Federal University of Pernambuco (UFPE)

IF867 – Introduction to Deep Learning

2024 – 2026

– Head Teaching Assistant

2025

– Teaching Assistant

2024

IF866 – Computational Creativity

2024 – 2025

– Teaching Assistant

Skills & Tools

Programming – Python, R, SQL, $\LaTeX$, Bash, Git.

Data Science & ML – PyTorch (Lightning), TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, OpenCV, MLFlow, Optuna, Weights&Biases.

SWE & DevOps – FastAPI, PostgreSQL, Redis, Docker, AWS (Lambda, Cognito), MinIO, Celery.

Languages – Portuguese (native), English.